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Roll No.

TEE-610

**B. TECH. (EEE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

INDUSTRIAL INSTRUMENTATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are load cells ? Describe the working of a magnetoelastic load cell in detail. List the various applications of load cells. (CO1)
- (b) Give suitable comparison between hydraulic, pneumatic and strain gauge type load cell. (CO1)
- (c) Explain the relative angular twist method for torque measurement. (CO1)
2. (a) Explain, how the torque is measured. What are the various types of dynamometers available for torque measurement ? Explain any *one* in brief. (CO2)

P. T. O.

(2)

- (b) Explain the principle and working of a Drag cup type tacho. (CO2)
- (c) Elucidate the operation of AC and DC tacho generator. (CO2)
- 3. (a) What is Seismic Instrument ? Mention the quantities required to be measured in vibrating system. Give different modes of Seismic instruments. (CO3, CO4)
- (b) Explain the working of Saybolt viscometer and Rotameter type viscometer. (CO3, CO4)
- (c) What are the various methods to measure moisture ? Explain any *one* technique in detail. (CO3, CO4)
- 4. (a) List the various standards for temperature measurement accepted globally. What do you understand by Primary and Secondary fixed points ? (CO5)
- (b) List the various types of filled in system thermometers. Explain the working of any *one* in detail. (CO5)
- (c) What is a Thermocouple ? Describe the laws of thermocouple. Explain, how temperature is measured using thermocouple. (CO5)
- 5. (a) Explain the working of an elastic type pressure gauge. (CO6)
- (b) List the different methods of pressure measurement where electricity is involved. Explain any *one* such method in detail. (CO6)
- (c) Explain, in detail, the working of a capacitive type pressure gauge. (CO6)

Roll No.

TEE-606

B. TECH. (EEE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022

COMPUTER ARCHITECTURE**Time : Three Hours****Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) In how many functional parts the microinstruction format is divided ?
Explain the function of each field. (CO1)
- (b) Discuss the differences between hardwired control unit and microprogrammed control unit with the help of block diagrams. (CO1)
- (c) Explain the algorithm for floating point addition and subtraction along with an example. (CO1)

P. T. O.

2. (a) What is microoperation ? What are the various types of microoperation ?
(CO2)
- (b) Explain two level and three level memory organization for both independent and memory hierarchy. What will be the total average time in all the cases ?
(CO2)
- (c) What are magnetic disk and optical disk ? Write down the differences between these two.
(CO2)
3. (a) What is I/O interface ? Why is it needed in a computer system ? Explain along with an example.
(CO3)
- (b) What is Bus Arbitration ? How many mechanisms are there for bus arbitration ? Explain any *one* along with its advantages and disadvantages.
(CO3)
- (c) What is a port in computer system ? Explain parallel port with the help of a diagram.
(CO3)
4. (a) Explain all the registers mentioned below in terms of their size and use :
(CO4, CO5)
- (i) RIP and RFLAGS
 - (ii) XMM Registers
 - (iii) YMM Registers
 - (iv) FPU Registers
- (b) Discuss some architectural features of 80 × 86, IA - 32 microprocessors.
(CO4, CO5)
- (c) What do you mean by BIU and EU ? How are they connected ?
(CO4, CO5)

(3)

5. (a) What is pipelining ? Write the advantages and disadvantages of pipelining. (CO5, CO6)
- (b) What do you mean by data hazards and pipelining hazards ? How many types of pipelining hazards are there ? Explain any *one*. (CO5, CO6)
- (c) Define the following : (CO5, CO6)
- (i) Dynamic Scheduling
 - (ii) Dependability

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Roll No.

TEE-603

**B. TECH. (EEE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

POWER SYSTEM PROTECTION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Differentiate between the following : (CO1)
 - (i) Switch and Switchgear
 - (ii) Fuse and Circuit Breaker
- (b) Discuss the concept of dividing the system into protective zones. Draw suitable diagram to support the explanation. (CO1)
- (c) Analyze the transient behavior in Synchronous machine and discuss how overall reactance increases with time. (CO1)
2. (a) What are the fundamental requirements of protective relaying ? (CO2)

P. T. O.

- (b) The current setting is set at 75%, TMS = 0.4; C. T. ratio = 400/5, Fault current is 6000 A. Determine the operating time of the relay. At TMS = 1 the operating time at various PSMs are shown in the table below given : (CO2)

PSM	Operating time (in secs)
2	10
4	5
5	4
8	3
10	2.8
20	2.4

- (c) Explain the distance relay protection scheme. How are they superior to overcurrent relays ? (CO2)
3. (a) What is the Buchholz relay ? Which equipment is protected by it ? For what type of fault is it employed ? With a neat diagram, discuss its working principle. (CO3)
- (b) Explain Merz Price system of protection as applied to alternator. (CO3)
- (c) Discuss the operating principle of electromagnetic induction relays. (CO3)
4. (a) What are the merits and demerits of static relay over electromagnetic relay ? (CO4, CO5)
- (b) Differentiate between making and breaking capacity of circuit breaker. Why asymmetrical breaking current is higher than symmetrical breaking current in a circuit breaker ? (CO4, CO5)

(3)

- (c) In a power system the r.m.s. voltage is 38.1 kV. It has inductance and capacitance of 10 mH and 0.02 μ F respectively. Determine :

(CO4, CO5)

- (i) the maximum voltage across the breaker contacts
- (ii) frequency of transient oscillation
- (iii) the average RRRV
- (iv) the maximum RRRV.

5. (a) Define restriking voltage. Derive an expression for restriking voltage in terms of system capacitance and inductance. (CO5, CO6)

- (b) Write a short note on SF₆ circuit breaker. (CO5, CO6)

- (c) Obtain rated symmetrical breaking current, making capacity, short-time current rating for a 3-phase oil circuit breaker having the following data : (CO5, CO6)

Operating voltage = 33 kV

Rupturing capacity = 700 MVA

Normal current = 500 A

Time = 3 sec

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TEE-602

**B. TECH. (EE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

POWER SYSTEM—II

Time : Three Hours

Maximum Marks : 100

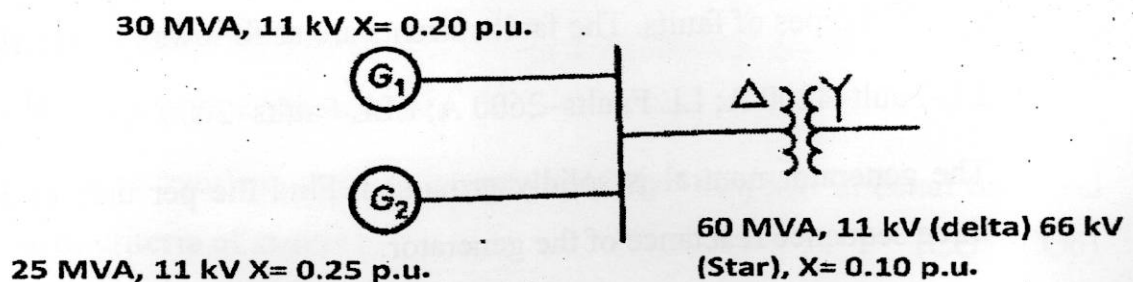
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by per unit representation of power system quantities ? Show that total reactance of the transformer refers to primary and secondary are equal if represented in p.u. (CO1)
- (b) Assuming S bases of 25 and 60 MVA, calculate the through impedance in ohms between the generators and the output terminals of the transformer for the system, as shown in Fig. given below. Specifications of the generators and transformer are indicated in figure. (CO1)



P. T. O.

- (c) A 5 kVA 400/200 V, 50 Hz single-phase transformer has primary and secondary leakage reactance each of 2.5Ω . Determine total reactance in p.u. (CO1)
2. (a) Draw zero sequence network of the transformer having (i) primary winding in star with neutral and secondary winding delta and (ii) both primary and secondary winding in delta. (CO2)
- (b) Discuss symmetrical component analysis of three-phase unbalanced system. Also, derive symmetrical components in terms of unbalanced voltage phasors. (CO2)
- (c) In a three-phase four wire system the currents in the lines a , b and c under abnormal conditions of loading were as follows : (CO2)

$$I_a = 50 \angle 45^\circ, I_b = 40 \angle 300^\circ \text{ and } I_c = 30 \angle 180^\circ$$

Calculate the zero, positive and negative phase sequence currents in line a and the return current in the neutral conductor.

3. (a) Discuss the various types of shunt faults existing in transmission systems. Also, draw the interconnection of sequence network in LLG faults conditions. (CO3)
- (b) A 50 MVA, 11 kV three-phase synchronous generator was subjected to different types of faults. The fault currents are as follows : (CO3)
- LG-Fault-4200 A; LL Faults-2600 A; LLL Faults-2000 A.

The generator neutral is solidly grounded. Find the per unit values of three sequence reactance of the generator.

- (c) A star connected three-phase synchronous generator has solidly grounded neutral and feeds a line upon which there is fault. The total impedances upto the fault are as follows : (CO3)

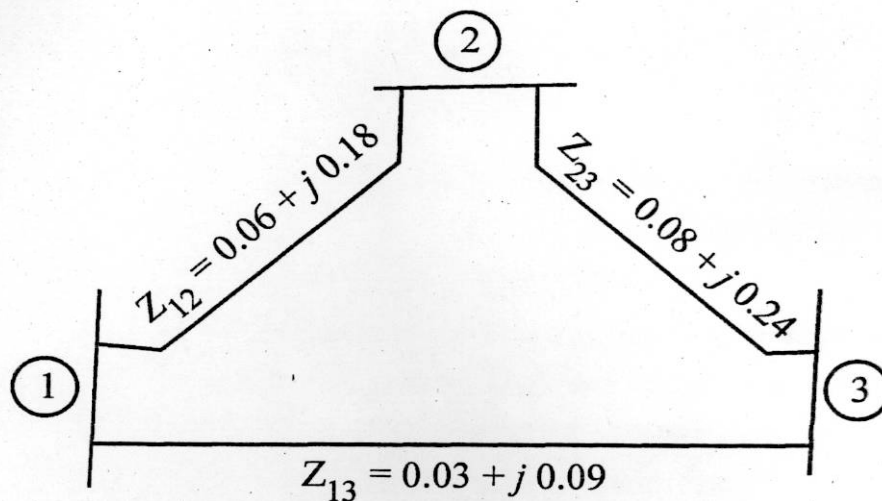
Positive sequence impedance : $j2.2 \Omega$

Negative sequence impedance : $j1.9 \Omega$

Zero sequence impedance : $j1.4 \Omega$

If the generator induced voltage is 11 kV, Calculate the fault current for
(i) an LG fault. (ii) a LL faults.

4. (a) Discuss the Gauss-Seidel method of load flow analysis. (CO4)
(b) Determine Y_{bus} for the 3-bus system as shown in Fig. The line series impedances between the buses are indicated in Fig. (CO4)



- (c) Derive static load flow equations and discuss the desirable feature of an ideal load flow analysis. (CO4)
 5. (a) What is transient and steady state stability ? Explain in detail the equal area criteria of stability. (CO5, CO6)

P. T. O.

- (b) Consider a synchronous generator supplies a power to the infinite bus through double circuit line. Derive the expression of critical clearing angle, using equal area criterion, when faults are existing in one of the transmission line. (CO5, CO6)
- (c) Define Short Circuit Capacity (SCC) of bus of a network. Derive the expression of SCC in terms of base VA and Thevenin's impedance of network in p.u. Also, write the procedure for calculating SCC. (CO5, CO6)

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Roll No.

TEE-601

**B. TECH. (EEE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

POWER ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the working of an IGBT and also draw its switching characteristics. (CO1)
- (b) Draw the circuit symbol and V-I characteristics of a TRIAC and explain its working. (CO1)
- (c) Explain the requirements of commutation circuit for thyristor and describe resonant pulse commutation circuit with appropriate waveforms. (CO1)

P. T. O.

2. (a) Draw the circuit diagram and output voltage waveforms of a three-phase fully-controlled bridge converter with highly inductive load for firing angle of 90 degrees. Derive an expression for average output voltage.
(CO2, CO6)
- (b) A single-phase full wave a.c. controller feeds a load of $R = 20 \Omega$ with an input voltage of 230 V, 50 Hz. Firing angle for both the thyristor is 45° , calculate (i) r.m.s. value of output voltage, (ii) load power and input power factor and (iii) average and r.m.s. current of thyristor.
(CO2, CO6)
- (c) Discuss the operation of single-phase a.c. regulator with RL load when firing angle is less than or equal to the load phase angle. Show that for this case, the output voltage of a.c. regulator cannot be regulated.
(CO2, CO6)
3. (a) Describe with neat circuit diagram and relevant wave forms the working of three-phase bridge inverter under 120° mode of conduction for a balanced delta connected resistive load. Also discuss the advantages of 120° over 180° conduction mode.
(CO3, CO6)
- (b) Discuss the circuit diagram of series inverter and indicate the need for an optimum time margin. Also indicate the merits and demerits of this inverter.
(CO3, CO6)
- (c) A single-phase full converter feeds power to RLE load with $R = 6 \Omega$, $L = 6 \text{ mH}$ and $E = 60 \text{ V}$. The source voltage is 230 V. For continuous conduction, find the average value of load current for firing angle of 50 degrees. In case one of the four SCRs gets open circuited, find the new value of average, current assuming continuous output current.
(CO3, CO6)

(3)

4. (a) Draw the circuit diagram of class E chopper and explain its four-quadrant operation in brief. (CO4, CO6)
- (b) The single-phase voltage controller has input voltage 230 V, 50 Hz and a load of $R = 15 \Omega$, for 6 cycle on and 4 cycle off. Determine :
- (i) r.m.s. output voltage
 - (ii) input power factor
 - (iii) average and r.m.s. thyristor current. (CO4, CO6)
- (c) Discuss the advantage of multi-pulse width modulation scheme over single pulse width modulation for voltage control of single-phase inverter. Draw the waveform for multiple pulse width modulation scheme. (CO4, CO6)
5. (a) Describe the basic principle of working of single-phase to single-phase step down cyclo converter with voltage and current waveforms for both continuous and discontinuous load conductions. (CO5, CO6)
- (b) Discuss with neat circuit diagram of three-phase to single-phase cyclo converter with RL load. Also draw the output voltage waveform. (CO5, CO6)
- (c) Explain the principle of on-off control and phase control. What are the advantages and disadvantages of both types of control ? (CO5, CO6)